

Egyptian rendering / XSR 0.10

01 / Signs and composition

Natural outline proportions, centered H/V groups and stable nesting.

Structure	Noto Sans EgyptHiero	NewGardiner	Segoe UI Historic
Single tall sign A1			
Single flat sign N35			
Horizontal A1 H N35			
Vertical A1 V N35			
Nested H over V (A1 H B1) V G17			
Nested V beside H A1 H (B1 V G17)			
Different proportions A1 H N35 H G17			

Inspection notes

Identical Unicode in every column. Frames show the computed layout extent. H = U+13431; V = U+13430. Parentheses describe the structure, not literal punctuation.

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02 / Outline-aware insertion

Each inset is placed in a measured empty region of the host outline.

Structure Noto Sans EgyptHiero NewGardiner Segoe UI Historic

Top-left / top-start

U+13432 / F4 + X1



Bottom-left / bottom-start

U+13433 / I10 + A1



Top-right / top-end

U+13434 / D17 + X1



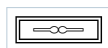
Bottom-right / bottom-end

U+13435 / G25 + X1



Middle / internal

U+13439 / N37 + 034



Top opening

U+1343A / D28 + J1



Bottom opening

U+1343B / D32 + V28



Inspection notes

Inset locations come from scan-converted outlines with a collision margin. Shapes with no usable cavity fail explicitly; no font-specific insertion coordinates or alternate glyph recipes are used.

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03 / Overlays and enclosures

Overlays share ink centers. Vector enclosures grow around actual content.

Structure	Noto Sans EgyptHiero	NewGardiner	Segoe UI Historic
Overlay D36 overlay V28			
Grouped overlay (A1 H B1) overlay (G17 V N35)			
Cartouche V11a / U+1343C..D / V11b			
Rectangular enclosure 06a / U+1343C..D / 06d			
Walled enclosure 036a / U+1343E..F / 036b			
Missing enclosure ends Bare U+1343C..D			
Enclosed insertion Cartouche with F4 + X1			

Inspection notes

Common cartouche, rectangular and walled end pairs are drawn by XSR. Missing end delimiters leave that end open. Overlay overlap is intentional. Alternate endpoints use outlines from the selected font; common ends remain vectors.

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04 / Transforms, damage and combinations

Rotation precedes mirroring. Hatching marks damaged or missing text.

Structure	Noto Sans EgyptHiero	NewGardiner	Segoe UI Historic
Mirror A1 / U+13440			
Quarter turn V26 / VS1 / 90 degrees clockwise			
Diagonal turn F51 / VS4 / 45 degrees clockwise			
Rotate, then mirror F16 / VS1 / U+13440			
Partial damage A1 / U+1344B / top half			
Full damage G17 / U+13455			
Lost signs Full / half / tall / wide			
Nested insertion F4 top-left (D28 top J1)			
Insertion plus H/V I10 bottom-left A1, beside G17			

Inspection notes

Normal compatible outline fonts use one code path. Rotation selectors are applied geometrically, not as font-specific variants. Only registered sign + selector pairs rotate. Unregistered pairs are diagnosed explicitly.

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05 / Standardized rotation and horizontal RTL

The same selector means different angles for different signs. RTL reverses logical start/end.

Structure Noto Sans EgyptHiero NewGardiner Segoe UI Historic

VS4: 30 degrees A15 + FE03			
VS4: 25 degrees H6 + FE03			
VS4: 40 degrees F44 + FE03			
VS4: 15 degrees Aa11 + FE03			
LTR: A1, B1, G17 Logical order, separate quadrats			
RTL: A1, B1, G17 Read from the right			
RTL top-start Insertion is now on the right			
RTL top-end Insertion is now on the left			
RTL rotation + mirror F16 / FE00 / U+13440			

Semantic policy

Unicode 18 standardized variants determine angles. Explicit NoMirror violations and unregistered NoRotate variants are errors. Registered angle-specific variants take precedence over the broad NoRotate property. Full vertical text flow is reserved; V JOINER within a quadrat remains supported.

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06 / Editorial marks, continuous loss and line wrapping

Brackets use explicit Egyptian context; ordinary punctuation retains its normal TeX meaning.

Structure Noto Sans EgyptHiero NewGardiner Segoe UI Historic

Square / partial brackets

Destruction and partial
destruction



Angle / double brackets

Addition and ancient deletion



Editorial braces

Deletion, within explicit
context



Continuous lost signs

Lost + FE00 / adjoining
regions



Damaged start, open end

Cartouche cap + damage



RTL open end

Logical opening is on the
right



Alternate endpoints

06b / 06f, selected-font
outlines



Paragraph wrapping / atomic H quadrats

Noto Sans EgyptHiero



RTL (same logical input)



NewGardiner



RTL (same logical input)



Segoe UI Historic



RTL (same logical input)



Inspection notes

TeX can break between quadrats; the A1–B1 H group stays together. Expanded shading meets without white gaps. NoMirror prohibits explicit mirroring that changes identity; RTL orientation is independent. Contextual GSUB alternates remain a future font-quality enhancement.